

Centre of Mass, Linear Momentum and Collisions, **solved**

Step-by-step solutions to every practice problem in Chapter 6. Try each problem on your own before you read its solution.

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P6.1

Yes. The centre of mass is a mass-weighted average position, and it need not coincide with any material point. Examples:

- **A ring:** the CM is at the empty centre.
- **A horseshoe or a boomerang:** the CM is in the space between the arms.

The Fosbury flop. The jumper goes over the bar head-first and face-up, arching the back strongly. At the moment the hips pass over the bar, the head and shoulders have already dropped down on the far side, and the legs still hang down on the near side. Like the ends of a horseshoe draped over a rail, enough mass hangs below the bar on both sides that the CM passes **below** the bar, even though each part of the body passes over it. The jumper therefore clears a bar higher than the peak height of their CM. This saves energy compared with the older “straddle” technique.

P6.2

The impulse–momentum theorem says $F_{\text{avg}}\Delta t = \Delta p$. When you land, or when an egg lands, Δp is fixed by the speed of impact: the body must be brought from mv to zero.

Bending your knees lets your body decelerate over a longer distance, and so over a longer time Δt . For the same Δp , a longer Δt means a proportionally smaller average force on your bones and joints (Example 6.6).

The egg. A pillow deforms over several centimetres, stretching the stopping time to tens of milliseconds. A hard floor stops the egg in a fraction of a millisecond, so the force is perhaps a hundred times larger and exceeds what the shell can withstand.

P6.3

No. Momentum must be conserved, and the bodies end up moving together at the CM velocity $V = P/M$. The kinetic energy

$$K_{\text{cm}} = \frac{1}{2}MV^2 = \frac{P^2}{2M}$$

must remain. What is lost is all the kinetic energy of motion *relative to* the centre of mass. Only when the total momentum is zero, as in two equal cars colliding head-on at equal speeds, is all the kinetic energy lost.

P6.4

Throw the wrench away from the spacecraft. The astronaut and the wrench start with zero total momentum, and no external force acts. After the throw, $m_a v_a = -m_w v_w$, so she moves *towards* the spacecraft. For example, a 2 kg wrench thrown at 5 m/s gives a 100 kg astronaut 0.1 m/s, and she reaches the craft in about 100 s.

Swimming motions cannot work. Arm and leg motions involve only internal forces, so the CM of the astronaut cannot move. She can change her orientation by wriggling, but she cannot change her position.

P6.5

Treat each square as a particle at its own centre. All three have equal mass m .

- Square 1: centre (5, 5)
- Square 2: centre (15, 5)
- Square 3: centre (5, 15)

$$x_{\text{cm}} = \frac{5 + 15 + 5}{3} = 8.33 \text{ cm}, \quad y_{\text{cm}} = \frac{5 + 5 + 15}{3} = 8.33 \text{ cm}.$$

The point **(8.33, 8.33)** lies inside the corner square ($0 \leq x, y \leq 10$), so **yes, it is on the plate**. By symmetry about the line $y = x$, it had to lie on that line.

P6.6

Put the origin at the Earth's centre:

$$x_{\text{cm}} = \frac{M_M d}{M_E + M_M} = \frac{(7.35 \times 10^{22})(3.84 \times 10^8)}{6.04 \times 10^{24}} = 4.67 \times 10^6 \text{ m.}$$

This is **inside the Earth**, about 4670 km from the centre, or 73% of the way to the surface.

Meaning. The Earth and Moon both orbit this point, the *barycentre*, once a month. So the Earth's centre traces a small circle of radius 4670 km each month. Astronomers use exactly this kind of “wobble” to detect planets around other stars: the star moves around the star-planet barycentre, and its light shows a periodic Doppler shift.

P6.7

Take the rebound direction as positive, so $v_i = -12$ m/s and $v_f = +10$ m/s.

Impulse:

$$J = m(v_f - v_i) = 0.40(10 + 12) = 8.8 \text{ N s (away from the wall).}$$

Average force:

$$F = \frac{J}{\Delta t} = \frac{8.8}{0.020} = 440 \text{ N.}$$

Kinetic energy lost:

$$\frac{K_i - K_f}{K_i} = \frac{12^2 - 10^2}{12^2} = \frac{44}{144} = 0.31.$$

About 31% is lost.

P6.8

Impulse = area under the triangular F - t graph:

$$J = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}(0.010 \text{ s})(600 \text{ N}) = 3.0 \text{ N s}.$$

Speed, starting from rest:

$$v = \frac{J}{m} = \frac{3.0}{0.058} = 52 \text{ m/s (186 km/h)}.$$

This is typical of a hard serve.

P6.9

Take right as positive. Momentum is conserved:

$$2.0(3.0) + 1.0(-2.0) = 3.0 V \Rightarrow V = \frac{4.0}{3.0} = 1.33 \text{ m/s (to the right)}.$$

Kinetic energy:

$$K_i = \frac{1}{2}(2.0)(9.0) + \frac{1}{2}(1.0)(4.0) = 9.0 + 2.0 = 11.0 \text{ J}, \quad K_f = \frac{1}{2}(3.0)(1.33)^2 = 2.67 \text{ J}.$$

Kinetic energy lost: $11.0 - 2.67 = 8.3 \text{ J}$, about 76%.

P6.10

Using the elastic-collision formulas with $m_1 = 1.0 \text{ kg}$, $m_2 = 2.0 \text{ kg}$ and $v_1 = 6.0 \text{ m/s}$:

$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1 = \frac{-1.0}{3.0} (6.0) = -2.0 \text{ m/s}, \quad v'_2 = \frac{2m_1}{m_1 + m_2} v_1 = \frac{2.0}{3.0} (6.0) = 4.0 \text{ m/s}.$$

The lighter ball bounces back at 2.0 m/s, and the heavier one moves forward at 4.0 m/s.

Checks.

- Momentum: $1.0(6.0) = 1.0(-2.0) + 2.0(4.0) = 6.0$. ✓
- Kinetic energy: $\frac{1}{2}(1.0)(36) = 18 \text{ J}$ before, and $\frac{1}{2}(1.0)(4) + \frac{1}{2}(2.0)(16) = 2 + 16 = 18 \text{ J}$ after. ✓
- Relative velocity reverses: approach at 6.0 m/s, separation at $4.0 - (-2.0) = 6.0 \text{ m/s}$, so $e = 1$. ✓

P6.11

(a) **Recoil.** The heart's push on the blood is an internal force of the patient-plus-blood system, and the table exerts no horizontal force. So total momentum stays zero:

$$m_{\text{blood}}v_{\text{blood}} + M_{\text{body}}V = 0 \Rightarrow V = -\frac{0.070(1.0)}{70} = -1.0 \times 10^{-3} \text{ m/s}.$$

The body recoils at about 1 mm/s towards the feet.

(b) **Average force:**

$$F = \frac{\Delta p}{\Delta t} = \frac{0.070(1.0)}{0.10} = 0.70 \text{ N}.$$

(c) **Why it is hard to measure.** The body is about 1000 times heavier than the ejected blood, so the recoil velocity and displacement (a fraction of a millimetre) are tiny. The force, 0.7 N, is far smaller than the static friction between a 70 kg body and an ordinary mattress, which is hundreds of newtons. On an ordinary bed the body would simply not move. Early ballistocardiographs therefore used suspended or air-bearing tables. Modern devices use extremely sensitive accelerometers or force sensors and filter out the noise.

P6.12

(a) **Recoil speed:** $v_r = \frac{0.010(800)}{4.0} = 2.0 \text{ m/s}.$

Kinetic energies:

$$K_r = \frac{1}{2}(4.0)(2.0)^2 = 8.0 \text{ J}, \quad K_b = \frac{1}{2}(0.010)(800)^2 = 3200 \text{ J}.$$

The ratio $K_r/K_b = 1/400 = m_b/m_r$. Because the two momenta are equal in size, $K = p^2/2m$ shares the energy in inverse proportion to the masses.

(b) **Force on the shoulder:** the shoulder must remove 8.0 J over 5.0 cm:

$$F_{\text{avg}} = \frac{K_r}{d} = \frac{8.0}{0.050} = 160 \text{ N}.$$

(c) **Hold it firmly.** Held tight, the rifle and the shooter's upper body (say 30 kg in all) recoil together. The same momentum, 8.0 kg m/s, then gives a recoil energy of $p^2/2M = 64/60 \approx 1 \text{ J}$, much less than 8 J, and it is absorbed gradually. Held a few centimetres away, the rifle alone picks up the full 8 J, then *slams* into the shoulder, and that energy must be absorbed over a much shorter distance. The result is a painful and potentially injurious blow.

P6.13

(a) **Head-on collisions with carbon.** Each collision leaves the neutron with a fraction **0.716** of its energy (Example 6.8). After n collisions:

$$E_n = E_0(0.716)^n.$$

$$n = \frac{\ln(E_n/E_0)}{\ln 0.716} = \frac{\ln(0.025/2.0 \times 10^6)}{\ln 0.716} = \frac{\ln(1.25 \times 10^{-8})}{-0.334} = \frac{-18.2}{-0.334} = 55 \text{ collisions.}$$

(b) **Random angles.** Now $E_n = E_0 e^{-n\xi}$, so $n = \ln(E_0/E_n)/\xi = 18.2/\xi$:

- Graphite: $n = 18.2/0.158 \approx 115$ collisions.
- Water: $n = 18.2/1.0 \approx 18$ collisions.

Evaluate. Hydrogen moderates about six times faster than carbon. That is why light-water reactors can be compact, while graphite-moderated reactors need large moderator volumes. Glancing collisions transfer less energy than head-on ones, so realistic collision numbers are about twice the head-on estimate.

P6.14

(a) **Thrust.** The fuel burn rate is $|dm/dt| = 40\,000/100 = 400$ kg/s, so

$$F = u \left| \frac{dm}{dt} \right| = 2500(400) = 1.0 \times 10^6 \text{ N.}$$

Acceleration at lift-off:

$$a_0 = \frac{F - m_0 g}{m_0} = \frac{1.0 \times 10^6 - 4.9 \times 10^5}{5.0 \times 10^4} = 10.2 \text{ m/s}^2.$$

(b) **Derivation with gravity.** Newton's second law for the rocket, with the thrust included, is

$$m \frac{dv}{dt} = -u \frac{dm}{dt} - mg.$$

Multiply by dt/m and integrate from $t = 0$ to the burnout time t_b :

$$\Delta v = u \ln \frac{m_0}{m_f} - gt_b. \checkmark$$

Numbers:

$$\Delta v = 2500 \ln \frac{50\,000}{10\,000} - 9.8(100) = 2500(1.609) - 980 = 4024 - 980 = 3040 \text{ m/s}.$$

(c) Just before burnout, $m = 10\,000 \text{ kg}$:

$$a = \frac{1.0 \times 10^6}{1.0 \times 10^4} - 9.8 = 100 - 9.8 = 90 \text{ m/s}^2 \approx 9.2 g.$$

Evaluate. The acceleration grows as the fuel burns away. Crewed rockets throttle down near burnout to keep the load on astronauts below about 4 g. The term gt_b is the *gravity loss*. It is why rockets burn hard and fast rather than gently.

P6.15

(a) **Force.** Each second, 20 kg of sand must be given a horizontal velocity of 2.0 m/s, so the belt must supply horizontal momentum at the rate

$$F = v \frac{dm}{dt} = 2.0(20) = 40 \text{ N}.$$

(b) **Power:** $P = Fv = 40(2.0) = 80 \text{ W}$.

(c) **Rate of gain of kinetic energy:**

$$\frac{dK}{dt} = \frac{1}{2} v^2 \frac{dm}{dt} = \frac{1}{2} (4.0)(20) = 40 \text{ W}.$$

The missing 40 W. Each grain lands with no horizontal velocity and *slides* on the belt until it reaches the belt's speed. Kinetic friction during that sliding converts energy into heat. Exactly half of the motor's power is always lost this way, whatever the speed or the friction coefficient. This is the same 50% loss found in charging a capacitor from a battery (Chapter 17).

P6.16

Momentum (starting from rest): $1.0 v_1 = 3.0 v_2$, so $v_1 = 3v_2$, in opposite directions.

Energy (all 24 J of spring energy becomes kinetic energy):

$$\frac{1}{2} (1.0)(3v_2)^2 + \frac{1}{2} (3.0)v_2^2 = 4.5v_2^2 + 1.5v_2^2 = 6.0v_2^2 = 24 \Rightarrow v_2 = 2.0 \text{ m/s}, \quad v_1 = 6.0 \text{ m/s}.$$

Energy shares. The 1.0 kg block gets $\frac{1}{2}(1.0)(36) = 18$ J (75%), and the 3.0 kg block gets 6 J (25%). The lighter body takes the larger share of the energy, in inverse ratio to the masses, just like the bullet and the rifle.

P6.17

(a) Just before a bounce the ball has speed v , and just after it has ev . Since the height reached is $h = v^2/2g$, each rebound height is e^2 times the previous one: $h_n = e^{2n}h$.

(b) **Distance.** The ball falls h , then goes up and down each rebound height:

$$D = h + 2(e^2h + e^4h + \dots) = h + \frac{2e^2h}{1 - e^2} = h \frac{1 + e^2}{1 - e^2}.$$

Time. The first fall takes $t_0 = \sqrt{2h/g}$. A rebound with launch speed e^nv_0 lasts $2e^nv_0/g = 2e^nt_0$. So

$$T = t_0 + 2t_0(e + e^2 + \dots) = t_0 \left(1 + \frac{2e}{1 - e}\right) = \sqrt{\frac{2h}{g}} \frac{1 + e}{1 - e}.$$

Infinitely many bounces happen in a *finite* time, a real-life convergent geometric series.

(c) With $h = 10$ m and $e = 0.80$:

$$D = 10 \cdot \frac{1.64}{0.36} = 45.6 \text{ m}, \quad T = 1.43 \cdot \frac{1.8}{0.2} = 12.9 \text{ s}.$$

P6.18

Momentum is conserved, starting from zero. Take east as $+x$ and north as $+y$:

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = \vec{0}.$$

$$\vec{p}_1 = 1.0(30) \hat{j} = 30 \hat{j}, \quad \vec{p}_2 = 2.0(20) \hat{i} = 40 \hat{i} \quad (\text{kg m/s}).$$

$$\vec{p}_3 = -40 \hat{i} - 30 \hat{j} \Rightarrow \vec{v}_3 = \frac{\vec{p}_3}{3.0} = -13.3 \hat{i} - 10.0 \hat{j} \text{ m/s}.$$

Magnitude and direction:

$$v_3 = \sqrt{13.3^2 + 10.0^2} = 16.7 \text{ m/s}, \quad \tan^{-1} \frac{10.0}{13.3} = 36.9^\circ \text{ south of west}.$$

Energy released = total kinetic energy afterwards:

$$K = \frac{1}{2}(1.0)(30)^2 + \frac{1}{2}(2.0)(20)^2 + \frac{1}{2}(3.0)(16.7)^2 = 450 + 400 + 417 = 1.27 \times 10^3 \text{ J.}$$

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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